LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Microorganisms – Friend and Foe

Chapter / Topic: Microorganisms (topic-based)

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SECTION A

Q1. Malaria is caused by: **[1]**

Answer: (d) a protozoan (Plasmodium)

Q2. Yeast is used in making: **[1]**

Answer: (b) bread

Q3. Which of these diseases is caused by a virus? **[1]**

Answer: (a) Polio

Q4. The nitrogen-fixing bacterium found in the root nodules of leguminous plants is: **[1]**

Answer: (c) Rhizobium

Q5. Penicillin was discovered by: **[1]**

Answer: (b) Alexander Fleming

Q6. Salt helps preserve food because it: **[1]**

Answer: (a) draws out water, so microorganisms cannot grow

SECTION B

Q7. What are pathogens? Name one disease spread through contaminated water and one spread through air. **[2]**

Answer: Pathogens are disease-causing microorganisms. Water: cholera or typhoid. Air: tuberculosis or the common cold.

- Definition: 1
- Two diseases: ½ + ½

Q8. How does Lactobacillus change milk into curd? **[2]**

Answer: Lactobacillus multiplies in milk and converts the milk sugar into lactic acid. This sets the milk into curd.

- Multiplies in milk: 1
- Produces acid / sets curd: 1

Q9. What is fermentation? Who discovered it? **[2]**

Answer: Fermentation is the conversion of sugar into alcohol by yeast. It was discovered by Louis Pasteur in 1857.

- Definition: 1
- Louis Pasteur: 1

Q10. How can microorganisms cause food poisoning? Give one way to prevent it. **[2]**

Answer: Some microorganisms grow on food and produce toxic (poisonous) substances, so eating the spoilt food makes us sick. Prevention (any one): refrigerate food, cook it properly, cover it, eat it fresh.

- Toxins in spoilt food: 1
- Prevention: 1

SECTION C

Q11. Name the type of microorganism (bacterium or virus) that causes: **[3]**

(i) A bacterium (Bacillus anthracis). [1]
(ii) A virus. [1]
(iii) A bacterium. [1]

Q12. How does a vaccine protect us from a disease? Who discovered the vaccine for smallpox? **[3]**

Answer: A vaccine contains dead or weakened microbes. The body makes antibodies to fight them and remembers how to make them, so if the real disease-causing microbe enters later, it is destroyed quickly. Edward Jenner discovered the smallpox vaccine in 1798.

- Dead/weakened microbes: 1
- Antibodies + memory: 1
- Edward Jenner: 1

SECTION D

Q13. Answer the following: **[5]**

(i) (i) Decomposers break down dead plants, animals and organic waste into simple substances, keeping the environment clean. (ii) Bacteria such as Rhizobium, and blue-green algae, fix nitrogen and increase soil fertility. (iii) Antibiotics (e.g. penicillin) and vaccines are made from microorganisms. [3]

- Each: 1

(ii) Mosquitoes breed in stagnant water. Mosquitoes spread diseases such as malaria (and dengue), so removing standing water stops them from breeding. [2]

- Mosquitoes breed: 1
- They spread malaria/dengue: 1